

Michael Bau

Los Angeles, CA | (213) 995-3069 | mikebaumx@gmail.com

SUMMARY

Professional Senior Software Engineer with 10+ years of experience building scalable web applications across healthcare, ecommerce, and fintech. Proficient in TypeScript, Python, and Go with expertise in frontend development, backend services, API design, and cloud infrastructure. Experienced in AI/ML integration, microservices architecture, performance optimization, and test automation. Skilled at mentoring engineers, driving technical decisions, and collaborating cross-functionally to deliver high-impact products. Passionate about writing clean, well-tested code and improving developer experience.

EXPERIENCE

Senior Software Engineer

Headway | New York, NY | May 2023 – Present

- Developed patient portal with React, Next.js, TypeScript, and Tailwind CSS to create accessible search filters and booking flows, helping over a million patients find and schedule appointments with in-network therapists across the platform.
- Designed GraphQL schema and resolvers in Python with FastAPI framework and SQLAlchemy ORM to aggregate data from PostgreSQL, enabling clients to fetch nested provider and availability data in single requests across a platform serving 60,000+ therapists.
- Built caching layer using Go microservices that stored provider search results in Redis with TTL-based invalidation, reducing database load and improving response times during high-traffic periods when patient booking intent peaked.
- Implemented semantic search using RAG architecture where patient queries were converted to embeddings and matched against vector database representations of provider specialties, surfacing best-fit therapists and reducing abandonment.
- Engineered prompt templates and few-shot learning examples for LLM-based note generation in Python, iteratively tuning outputs to match clinical documentation standards and helping providers complete insurance-compliant progress notes in less time.
- Led technical design and cross-functional collaboration for AI feature integration, breaking down ambiguous product requirements into actionable engineering tasks and maintaining architecture decision records to document trade-offs as the roadmap evolved.
- Configured GitHub Actions pipelines building Docker images and deploying AI services to ECS using Fargate launch type in AWS, automating container orchestration and enabling zero-downtime deployments as LLM capabilities and prompt engineering patterns evolved.
- Built statically generated marketing pages and dynamic referral tracking using Next.js incremental static regeneration and server-side rendering for SEO metadata, improving organic search rankings to attract therapists in markets with limited provider coverage.
- Implemented experimentation framework using feature flags to control gradual rollouts and run A/B tests measuring user engagement across new features, enabling data-driven product decisions and safe iteration on high-impact changes before full deployment.
- Wrote Jest unit tests for React components, Playwright end-to-end automation for critical user flows, ensuring reliable releases and comprehensive test coverage across frontend components and backend services with confidence in production deployments.
- Set up monitoring dashboards in Datadog tracking API latency and error rates, configured Sentry for exception tracking with source maps, and responded to PagerDuty alerts during on-call rotation to maintain uptime for patients scheduling appointments.
- Conducted code reviews and pair programming sessions on Python backend services and TypeScript frontend features within two-week Scrum sprints, sharing patterns for error handling and state management to help engineers ship quality code confidently.

Full Stack Engineer

Apple | Cupertino, CA | September 2019 – April 2023

- Delivered premium shopping experience for millions of customers across 40+ countries by building responsive product catalog pages using React with TypeScript and SASS, ensuring consistent Apple brand standards and user experience across all regions.
- Built flexible GraphQL APIs on Node.js with Express.js backed by NoSQL databases to handle dynamic product configurations like custom engraving, storage options, and AppleCare plans while supporting rapid catalog updates during major product launches.
- Improved customer experience during major product releases by optimizing delivery of high-resolution imagery and videos through CDN configuration, which handled massive traffic spikes when new iPhone and Mac models launched with millions of visitors.
- Enhanced shopping personalization by integrating AI recommendation APIs built in Python that surfaced relevant accessories like AirPods, cases, and AppleCare based on customer device selections, which increased cross-sell opportunities throughout the purchase journey.

- Increased purchase confidence by building real-time inventory visibility through GraphQL queries with Redis caching across microservices, showing accurate stock levels across Apple retail stores globally for same-day pickup and online delivery options.
- Reduced checkout friction for premium device purchases by implementing express payment flows using React ES6 components with Apple Pay integration, enabling customers to complete iPhone and Mac purchases in seconds with saved payment and shipping information.
- Ensured reliable payment processing for high-value transactions by developing Node.js services to handle complex workflows including trade-in credits, gift cards, and financing options while maintaining strict PCI compliance standards to protect customer data.
- Maintained order fulfillment reliability during peak shopping periods like Black Friday and iPhone launch days by developing services using event-driven architecture with Apache Kafka that processed hundreds of thousands of orders asynchronously across regions.
- Improved post-purchase customer experience by connecting external CRM systems through REST APIs to automate order tracking and delivery updates, seamlessly coordinating between Apple's online orders, retail store pickup, and home delivery fulfillment channels.
- Maintained high quality standards for customer-facing features by implementing comprehensive test coverage using Jest for unit and integration testing, actively participating in Scrum ceremonies to plan sprints and ensure reliable releases ahead of launches.

Full Stack Developer

Morgan Stanley | New York, NY | November 2015 – August 2019

- Built portfolio dashboard for WealthDesk platform using Angular with RxJS for reactive data streams and TypeScript with Bootstrap components, consolidating client account data into single interface that enabled advisors to manage their entire book of business during client meetings.
- Developed RESTful backend services with Django and Python connected to PostgreSQL databases that processed portfolio calculations and handled thousands of daily transactions, maintaining data accuracy and system reliability for advisors serving high net worth clients.
- Migrated advisor-facing platform from Angular to React with Redux for centralized state management, improving code maintainability and enabling faster feature development for new WealthDesk capabilities that advisors requested to better serve their clients.
- Identified performance bottlenecks using New Relic monitoring and optimized Django backend by refactoring SQL queries, implementing Redis caching, and batching API calls, reducing dashboard load times for advisors during high-volume market periods.
- Collaborated with designers and product managers through Figma to build client reporting interface using React, TypeScript, and custom Sass styling, delivering polished UI/UX that enabled advisors to present quarterly portfolio reviews professionally.
- Designed scalable REST APIs with Python and Django that executed optimized SQL queries against PostgreSQL to deliver real-time risk analytics and portfolio insights, helping advisors identify investment opportunities and risks across client portfolios.
- Built risk analytics dashboard using React with Redux state management and Bootstrap components that visualized portfolio volatility and market scenario analysis, helping advisors communicate complex investment strategies and risk exposure to clients.
- Created comprehensive API documentation using Swagger and OpenAPI specifications for Django backend services, enabling internal development teams to integrate with WealthDesk platform efficiently and accelerating delivery of new advisor-facing features.
- Established full stack testing infrastructure with Jest for TypeScript frontend components and Pytest for backend services, ensuring platform reliability and catching regressions before production releases for advisors managing billions in client assets.
- Managed version control using Git and Azure DevOps while participating in Agile Scrum sprints with cross-functional teams, resolving merge conflicts through clear communication to maintain continuous delivery of WealthDesk platform updates without downtime.

EDUCATION

Bachelor of Science in Computer Science

University of Notre Dame | Notre Dame, IN | 2015

SKILLS

Programming Languages: TypeScript, Python, Go, HTML, CSS, JavaScript, Sass, SQL

Frontend Development: React, Next.js, Redux, Angular, RxJS, Vue.js, jQuery, Tailwind CSS, Bootstrap

Backend Development: Node.js, Express.js, Django, Flask, FastAPI, SQLAlchemy, GraphQL, REST APIs

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

Cloud & Infrastructure: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Docker, Kubernetes

AI/ML & Data: Large Language Model (LLM), Retrieval-Augmented Generation (RAG), Vector DBs, Prompt Engineering, OpenAI API, Anthropic API

Tools & Practices: CI/CD, GitHub Actions, Jest, Playwright, Apache Kafka, Salesforce, Datadog, Sentry, PagerDuty, New Relic, Agile, Scrum